Summer Industry Internship-I

On

BASIC CMOS COMPARATOR

Submitted in the partial fulfilment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

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UNDER THE GUIDANCE OF

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SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY

(Autonomous)

Yamnampet (V), Ghatkesar (M), Hyderabad – 501301.

2025-26

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CERTIFICATE

This is to certify that the Project report entitled "**BASIC CMOS COMPARATOR**" is being submitted by **Merugu.Vishwaksena, ROLL NO. 23311A0424** in partial fulfilment of the requirements for the award of Bachelor of Technology Degree in Electronics and Communication Engineering to Sreenidhi Institute of Science and Technology affiliated to Jawaharlal Nehru Technological University, Hyderabad (Telangana). This record is a bonafide work carried out by them under our guidance and supervision. The results embodied in the report have not been submitted to any other University or Institution for the award of any Degree or Diploma.

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on

Analog and Digital VLSI Circuit Design and Layout using CMOS and FinFET Technologies

Organized by the Department of ECE, Sreenidhi Institute of Science & Technology, Hyderabad,

held during 18th August to 15th September 2025, and submitted a project entitled

"Design of Basic CMOS Compartor using CMOS Technology".

This training is organized with the support of industry experts from Entuple Technologies, Bangalore, Intel Core Banglore and Mirafra Technologies-client AMD, Hyderabad.

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Convener

DECLARATION

I hereby declare that the work described in this project titled "**BASIC CMOS COMPARATOR"** which is being submitted by us in partial fulfilment for the award of Bachelor of Technology in the Department of **Electronics and Communication Engineering,** Sreenidhi Institute of Science and Technology is the result of investigations carried out by us under the guidance of **Dr. D. Asha Devi Professor, Dr. Sateesh Professor, Department of ECE, Sreenidhi Institute of Science and Technology, Hyderabad.**

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ACKNOWLEDGEMENTS

I would like to thank our Project Guide and Coordinator Dr. D. Asha Devi, Professor of ECE, Dr. Sateesh Professor summer industry internship coordinater for her technical guidance, constant encouragement and support in carrying out our project at college.

I profoundly thank Dr. S. P. V. Subba Rao, Head of the Dept. Electronics & Communication Engineering who has been an excellent guide and also a great source of inspiration to our work.

I would like to express our gratitude to all the industry persons for sharing their technical Cadence SW EDA – tools knowledge helped me to transform an idea into a real application.

I would like to express our heart-felt gratitude to our parents without whom I would not have been privileged to achieve and fulfill our dreams. We are grateful to our principal, Dr. T. Ch. Siva Reddy, who most ably run the institution and has had the major hand in enabling me to do our project.

The satisfaction and euphoria that accompany the successful completion of the task would be great but incomplete without the mention of the people who made it possible with their constant guidance and encouragement crowns all the efforts with success. In this context. We would like thank all the other staff members, both teaching and non-teaching, who have extended their timely help and eased our task.

Signature

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Abstract

A basic CMOS comparator is a digital or analog decision-making circuit that compares two input voltages (Vp and Vm) and produces a binary output, typically a logic 0 or 1. It functions by amplifying the differential input voltage to produce a high-gain output, often employing a pre-amplifier stage to condition the signal, a positive feedback or regenerative decision circuit for rapid switching, and an output buffer to drive a load. These circuits are vital in signal processing and data conversion, valued for their high integration, low power consumption, and ability to operate at high speed.

Keywords:

Input signals(Vin+

,Vin-)

Digital output

Pre-amplifier

Positive

feedback Latch

Gain stage

Output

buffer Tail

current

Input pair

Resolution

Power

consumption

Speed

Delay

hysteresis

CHAPTER 1: INTRODUCTION

1.1. A CMOS comparator is an electronic circuit using Complementary Metal-Oxide-Semiconductor (CMOS) technology that compares two analog input voltages and outputs a binary (high or low) digital signal representing the result of the comparison. If the positive input voltage is greater than the negative input voltage, the output is a logic one; if it is less, the output is a logic zero. CMOS comparators are crucial components in Analog-to-Digital Converters (ADCs) and other digital systems, offering advantages like low power consumption, high speed, and full-swing outputs.

1.2. If the positive input voltage is greater than the negative input voltage, the output is a logic one; if it is less, the output is a logic zero. CMOS comparators are crucial components in Analog-to-Digital Converters (ADCs) and other digital systems, offering advantages like low power consumption, high speed, and full-swing outputs.

CHAPTER 2: LITERATURE SURVEY

2.1. Input Pre-amplification Stage:

This stage amplifies and buffers the input analog signals, often using differential input transistors.

2.2. Regenerative or Decision Stage:

This critical stage uses positive feedback, often in the form of cross-coupled inverters, to convert the small, amplified voltage difference into a full-swing digital output.

2.3. CMOS comparators are widely used in:

Analog-to-Digital Converters (ADCs): They form the core of ADCs, converting analog signals into digital ones.

Data Receivers: Used in communication systems to interpret and process received dat

CHAPTER 3: METHODOLOGY

3.1 Library Creation

GPDK180nm Technology library is imported.

3.2 Schematic Design

The Basic CMOS comparator designed by using pmos and nmos transistors

3.3 Test Bench Design & Simulation Setup

To design a test bench for a CMOS comparator simulation, you need to create a separate piece of code or schematic that applies input signals (e.g., a sinusoidal wave, DC voltages) to the comparator circuit and drives a simulator (like Cadence, HSPICE, or Tanner) to verify the comparator's functionality and performance metrics, such as output voltage swing, power consumption, and delay.

3.4. Schematic -Based Test Bench: In Cadence Virtuoso, create a new schematic that instantiates your comparator circuit. You will add voltage sources, clock signals, and output probes to this test bench schematic.

3.5. Test Bench Setup: The document will describe the configuration of the test bench, including how input signals (e.g., sinusoidal waves, DC voltages) and a clock are applied to the comparator.

3.6. Theory

In electronics, a comparator is a device that compares two voltages or currents and outputs a digital signal indicating which is larger. It has two analog input terminals V_+ and V_- and one binary digital output V_o. The output is ideally

$$V_o = \begin{cases} 1, & \text{if } V_+ > V_-, \\ 0, & \text{if } V_+ < V_-. \end{cases}$$

A comparator consists of a specialized high-gain differential amplifier. They are commonly used in devices that measure and digitize analog signals, such as analog-to-digital converters (ADCs), as well as relaxation oscillators.

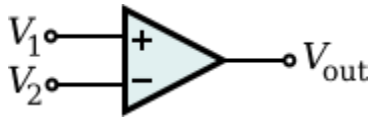
$V_{S-} \leq V_+, V_- \leq V_{S+},$

The differential voltages must stay within the limits specified by the manufacturer. Early integrated comparators, like the LM111 family, and certain high-speed comparators like the LM119 family, require differential voltage ranges substantially lower than the power-supply voltages (±15 V vs.36 V).[1] *Rail-to-rail* comparators allow any differential voltages within the power-supply range. When powered from a bipolar (dual rail) supply,or when powered from an unipolar TTL/CMOS power supply,

$0 \leq V_+, V_- \leq V_{cc}$.

Specific rail-to-rail comparators with p–n–p input transistors, like the LM139 family, allow the input potential to drop 0.3 volts *below* the negative supply rail, but do not allow it to rise above the positive rail.[2] Specific ultra-fast comparators, like the LMH7322, allow the input signal to swing below the negative rail *and* above the positive rail, although by a narrow margin of only 0.2 V.[3] Differential input voltage (the voltage between two inputs) of a modern rail-to-rail comparator is usually limited only by the full swing of power supply.

Op-amp voltage comparator



A simple op-amp comparator

An operational amplifier (op-amp) has a well balanced difference input and a very high gain. This parallels the characteristics of comparators and can be substituted in applications with low-performance requirements.[4]

A comparator circuit compares two voltages and outputs either a 1 (the voltage at the plus side) or a 0 (the voltage a the negative side) to indicate which is larger. Comparators are often used, for example, to check whether an input has reached some predetermined value. In most cases a comparator is implemented using a dedicated comparator IC, but op-amps may be used as an alternative.
Comparator diagrams and op-amp diagrams use the same symbols.

3.7.Design

A comparator consists of a high gain differential amplifier whose output is compatible with the logic gates used in the digital circuit. The gain is high enough that a very small difference between the input voltages will saturate the output, the output voltage will be in either the low logic voltage band or the high logic voltage band of the gate input. Analogue op amps have been used a comparators, however a dedicated comparator chip will generally be faster than a general-purpose operational amplifier used as a comparator, and may also contain additional features such as an accurate, internal reference voltage, adjustable hysteresis, and a clock gated input.

A dedicated voltage comparator chip such as LM339 is designed to interface with a digital logic interface (to a TTL or a CMOS). The output is a binary state often used to interface real world signals to digital circuitry (see analog-to- digital converter). If there is a fixed voltage source from, for example, a DC adjustable device in the signal path, a comparator is just the equivalent of a cascade of amplifiers. When the voltages are nearly equal, the output voltage will not fall into one of the logic levels, thus analog signals will enter the digital domain with unpredictable results. To make this range as small as possible, the amplifier cascade is high gain. The circuit consists of mainly bipolar transistors. For very high frequencies, the input impedance of the stages is low. This reduces the saturation of the slow, large p–n junction bipolar transistors that would otherwise lead to long recovery times. Fast small Schottky diodes, like those found in binary logic designs, improve the performance significantly though the performance still lags that of circuits with amplifiers using analog signals. Slew rate has no meaning for these devices. For applications in flash ADCs the distributed signal across eight ports matches the voltage and current

Speed: Measured by propagation delay, which indicates how quickly the comparator responds to input changes.

Power Dissipation: The amount of power consumed by the circuit, with a goal of minimizing it for battery-powered or high-density applications.

Offset Voltage: The small input voltage difference that causes the output to switch, with low offset being desirable to avoid errors.

Gain: The ability of the comparator to amplify the input voltage difference, which can be controlled in multi-stage designs.

Multi-Stage Design: Using two or more stages, often with a first stage providing an initial gain and a second stage for further amplification or to provide a delay before activation.

Leakage Transistor Techniques: Adding leakage transistors to the circuit's pull-up and pull-down networks to force more transistors OFF in the path from supply to ground, significantly reducing static power consumption.

3.7. Symbol

The standard symbol for a CMOS comparator is the same as an operational amplifier (op-amp) because it functions similarly as a high-gain differential amplifier. The process of creating this symbol involves abstracting the internal CMOS transistor-level circuitry into a simple, functional block representation for use in higher-level schematic design.

Symbol creation process in schematic capture tools

When using electronic design automation (EDA) tools like Cadence or Tanner, the process of creating a symbol from a complex CMOS circuit is a standard design flow procedure.

 Design the transistor-level schematic: First, the designer creates the full, detailed schematic of the CMOS comparator using PMOS and NMOS transistors. A common comparator architecture includes:

- Input preamplifier stage: Amplifies the small differential input signal.
- Regenerative latch stage: Uses positive feedback to rapidly drive the output to a logic high or low state.
- Output buffer stage: Converts the latch output into a full-scale digital logic signal.

 Define and label all ports: The designer must clearly define and label the connections that will be visible on the external symbol. These ports typically include:

- V_{in+}

$Vin+$

V_{in-}

$Vin-$

(analog inputs)

- V_{out}

$Vout$

(digital output)

- V_{DD}

VDD

(positive power supply)

-

V_{SS}

VSS

(ground)

- A clock signal (for dynamic comparators)

 Generate the symbol view: Most EDA tools offer a feature to automatically generate a basic symbol from a completed schematic. This creates a simple black-box representation with all the defined ports.

 Refine the symbol's appearance: For clarity and ease of use, the automatically generated symbol is typically refined and edited.

- Shape: The default square or box is replaced with the standard triangle shape.
- Pin placement: The pins for the inputs, output, and power are arranged logically.

 Save and use the symbol: Once the custom symbol is complete, it is saved to the design library. This allows the designer to use the simple triangular symbol in a higher-level schematic, treating the complex transistor-level design as single functional block. This process promotes hierarchical design, where complex circuits are built from reusable, abstracted components.

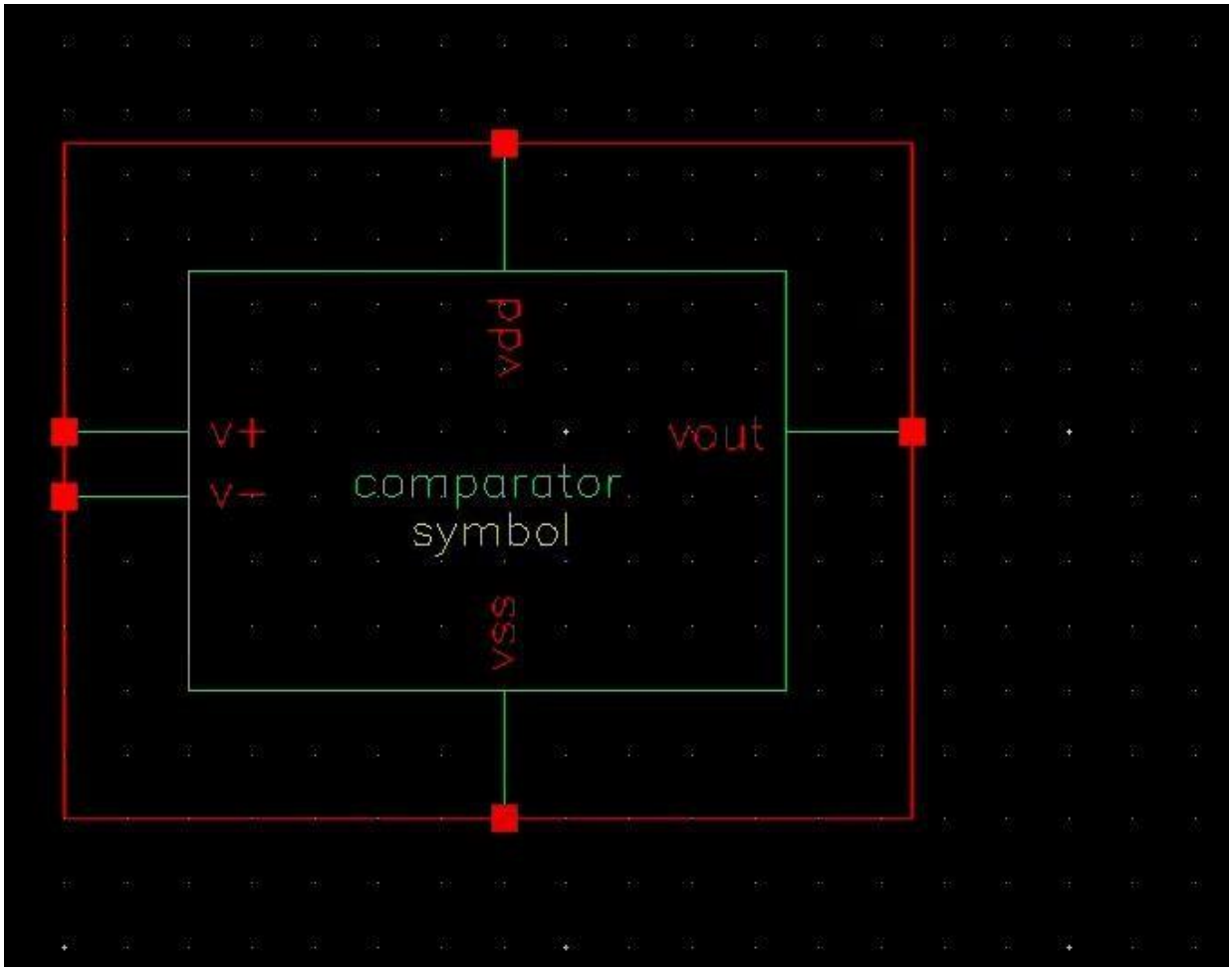


Figure 3.8

CHAPTER 4: IMPLEMENTATION

4.1 Schematic Design

A basic CMOS comparator compares two input voltages, with the output being a digital high or low signal indicating which input is greater. The schematic process involves a differential amplifier with NMOS and PMOS transistors, where the input voltages determine which transistors are conducting, leading to a change in the output voltage to one of the power rails. The core idea is that the comparator's structure amplifies the tiny difference between the input voltages to produce a full-scale digital output.

Schematic Process:

1. Differential Input Stage:

The comparator receives two input voltages, typically applied to the non-inverting and inverting terminals. These inputs drive a differential amplifier stage, which is the core of the comparator.

2. CMOS Transistor Action:

This differential stage uses complementary metal-oxide-semiconductor (CMOS) transistors (both NMOS and PMOS).

- When one input voltage is higher than the other, it turns on certain transistors and turns off others in the differential pair.
- For instance, if the non-inverting input voltage is higher, a specific set of transistors conducts, and the current path through the output stage changes.

3. Amplification and Output:

The output of the differential pair is then fed into subsequent amplification stages. Because of the CMOS structure, this amplification is very efficient, and the comparator converts the small differential voltage into a large output swing.

4. Digital Output:

The output voltage quickly swings to either the positive power supply rail (logic high) or the negative power supply rail (logic low), providing a digital signal that clearly shows which input voltage was larger.

How it Works:

- Input Voltage Comparison:

The comparator fundamentally functions by observing the difference between two input voltages.

- Switching Point:

The output switches between its high and low states when the two input voltages become equal, a point known as the switching threshold.

- Applications:

This process is crucial in applications like analog-to-digital converters (ADCs) and alarm circuits, where an analog voltage needs to be converted to a digital signal or a threshold needs to be detected.

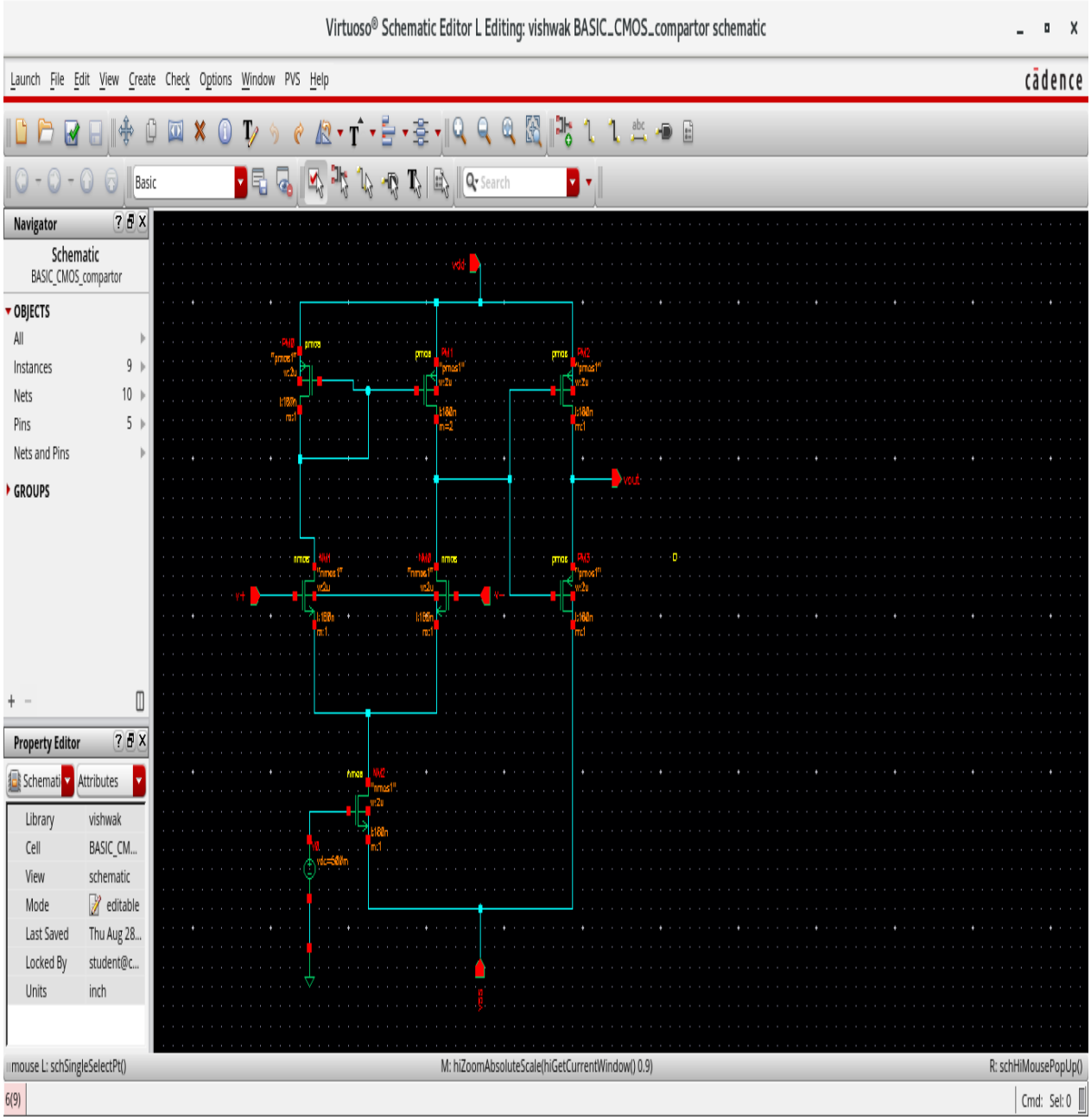


Figure 4.1

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Open collector output

Main article: Open collector

Some comparators (e.g. LM339) use open collector output to help interface to different logic families. When the inverting input is at a higher voltage than the non-inverting input, the output of the comparator connects to the negative power supply. When the non inverting input is higher than the inverting input, the output is high impedance, so the output voltage in this state can be set by an external pull-up resistor to a different voltage supply.

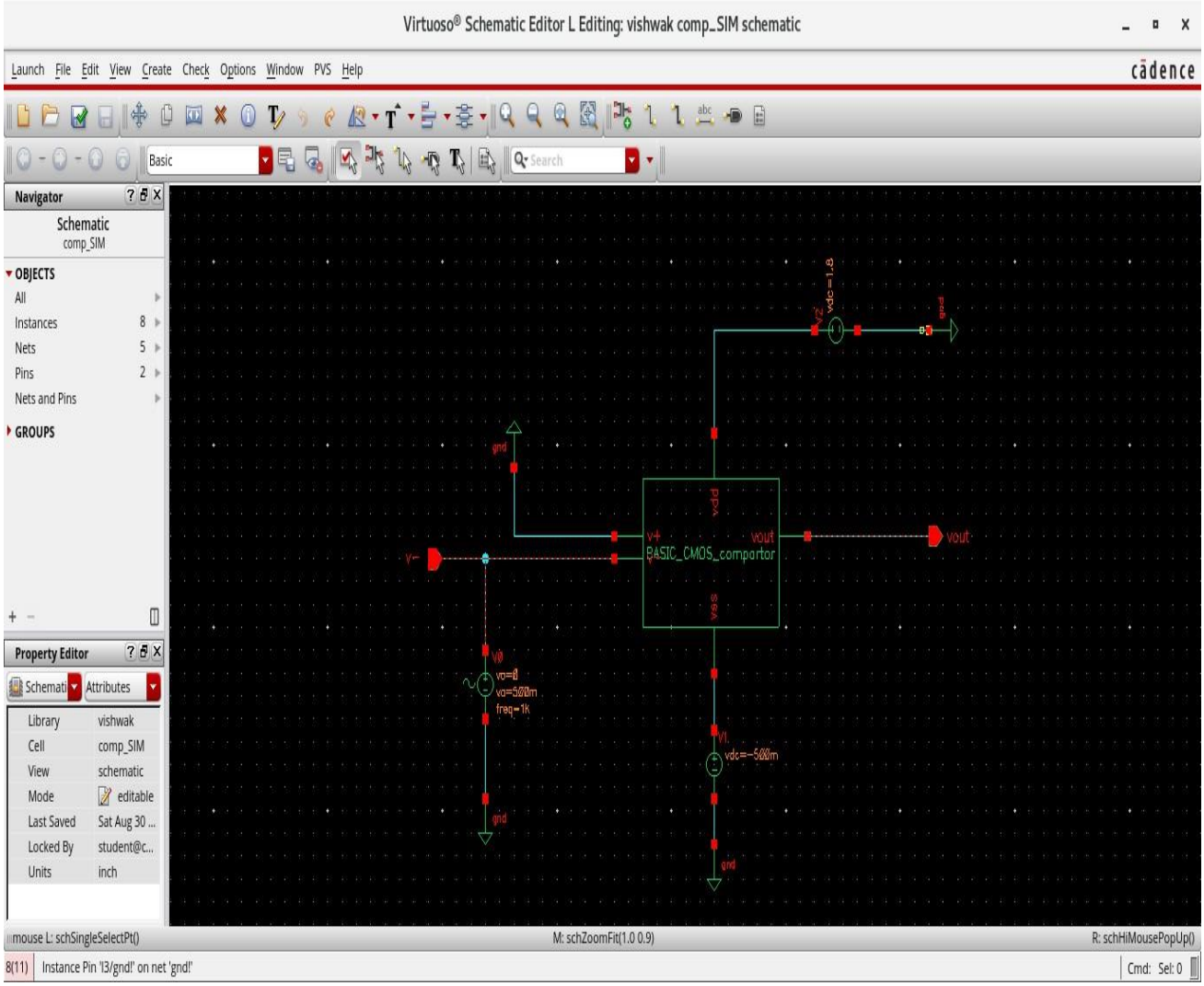

Figure 4.2

4.3. Simulation Tools: The paper will specify the software used, such as Cadence Virtuoso or Tanner EDA, and the CMOS technology process (e.g., 180nm, 0.5 micron).

4.4. Simulation Results: You will find figures showing the output waveforms (e.g., square wave for comparator) and tables summarizing key performance metri.

3. Simulation setup

- Testbench schematic: Show the test circuit used for simulations. This should include:

- The comparator circuit.
- Input voltage sources (V_{IN+} and V_{IN-}).
- Power supply connections (V_{DD} and V_{SS}).
- A capacitive load at the output.

- Simulation parameters: List the specific settings for each analysis, including:
  - Power supply voltages (e.g., V_{DD} = 1.8V, V_{SS} = 0V).
  - Input stimulus details (e.g., signal type, amplitude, frequency).
- Analysis types: Clearly state which simulations were performed.

2. DC sweep analysis

This analysis determines the comparator's static transfer characteristics, showing how the output switches based on the input voltage.

1. Inputs: Set one input (e.g., V_{IN-}) to a constant DC reference voltage (V_{REF}), typically half of the supply voltage.
2. Sweep: Create a DC sweep analysis for the other input (V_{IN+}). Sweep its voltage across the entire input common-mode range, for example, from (V_{SS} to V_{DD})
3. Outputs: Plot the output voltage (V_{OUT}) against the swept input voltage (V_{IN+}).
4. Results: The plot should show a very steep transition from one rail to the other. The voltage at which this sharp transition occurs is the trip point, and the steepness of the curve indicates the comparator's DC gain.

3. Transient analysis

This is the most critical simulation for assessing the comparator's dynamic performance, especially its speed.

Propagation delay measurement

1. Inputs: Apply a time-varying signal to the inputs. Use a small, fast-rising step input or a sine wave with one input fixed at V_{REF}

. The input difference signal (ΔV_{IN}) should be large enough to ensure a clean output transition.

2. Simulation: Run a transient simulation for a sufficient time period to capture the full input and output transitions.
3. Measurement:
   - t_{PLH}

(Propagation Delay, Low-to-High): Measure the time difference between the input crossing its reference voltage (V_{REF}) and the output crossing 50% of its final value.

   - t_{PHL}

(Propagation Delay, High-to-to-Low): Measure the time difference between the input crossing its reference and the output crossing 50% of its final value on the falling edge.

   - Total Propagation Delay (t_{PD}): Calculate the average:

$t_{PD} = (t_{PLH} + t_{PHL})/2$

tPD=($tPLH$+$tPHL$)/2

.

4. Analysis: The propagation delay is a key metric for determining the comparator's maximum operating frequency.

Power consumption measurement

1. Simulation: Use the same transient analysis as for the propagation delay.
2. Measurement: The simulator can calculate and plot the instantaneous power consumption. For a time-averaged result, measure the average power drawn from the power supply sources (V_{DD}) over a full comparison cycle.
3. Calculation: Some simulators provide an automatic power calculation function. Otherwise, you can measure the current from the supply and multiply it by the supply voltage. For dynamic comparators, the power depends on the clock frequency, load capacitance, and supply voltage.

4. Monte Carlo analysis

This simulation quantifies the impact of random process variations on the comparator(V_{OS}).

1. Setup: Define the analysis as a Monte Carlo simulation. Specify the number of runs (e.g., 100 or 500). For each run, the simulator will randomly vary device parameters (like threshold voltage and transistor width/length) within statistical limits defined by the process model.
2. Analysis Type: Run a DC sweep or transient analysis for each Monte Carlo iteration to find the trip point.
3. Measure Offset: For each run, the input-referred offset voltage (V_{OS}) is the difference between the actual trip point and the ideal reference voltage (V_{REF}).
4. Results: After all runs are complete, the simulator will present a histogram of the offset voltage distribution.

- Mean: The mean of the distribution indicates any systematic mismatch in the layout or design.
- Standard Deviation (σ): The standard deviation is a measure of the random offset caused by process variations.

5. Output: Report the mean and standard deviation of the offset voltage.

5. Corner analysis

For robustness, designers check performance under different process, voltage, and temperature (PVT) corners to ensure the design works reliably under extreme conditions.

1. Process Corners: Simulate the comparator using different transistor models:
   - TT (Typical): Typical NMOS and PMOS behavior.
   - FF (Fast-Fast): Both NMOS and PMOS are fast.
   - SS (Slow-Slow): Both NMOS and PMOS are slow.
2. Voltage Corners: Vary the supply voltage (VDD) to its specified maximum and minimum extremes.
3. Temperature Corners: Change the simulation temperature to the specified operating range (e.g., $-40\circ C$to $125\circ C$).
4. Verification: For each corner combination (e.g., TT corner at max $VDD$and high temp), run the DC and transient simulations to verify that the comparator's performance remains within specifications.

CHAPTER 5: RESULT

5.1. Analysis: The results are discussed in terms of speed (e.g., delay), power consumption, input-referred noise, and comparisons to previous work or other comparator types.

- **DC gain and offset:** Ensure the gain is high enough to produce a sharp transition, and that the offset is within the acceptable range for the target application (e.g., less than half the LSB for an ADC).
- **Propagation delay and speed:** Verify that the propagation delay meets the timing requirements for the application, such as the maximum sampling rate.
- **Power consumption:** Check that the power dissipation is within the budget, which is especially critical for low-power and portable devices.
- **Mismatch and robustness:** Analyze the Monte Carlo results to ensure that the offset voltage distribution is tight and that the circuit is robust against manufacturing variations.

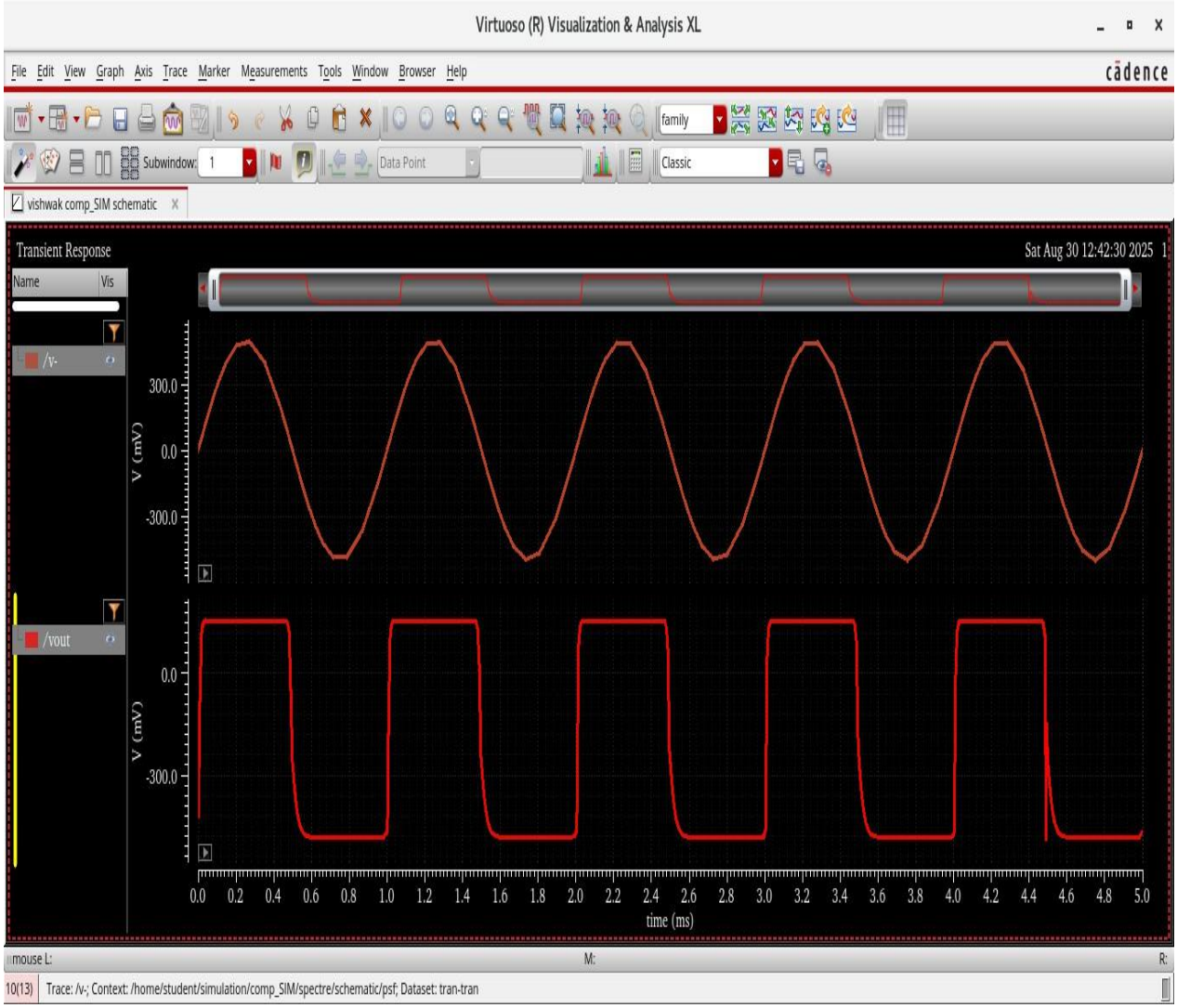

Figure 5.1

References:-

1]. *Design of Analog CMOS Integrated Circuits* by Behzad Razavi:

2]. *CMOS Analog Circuit Design* by Phillip E. Allen and Douglas R. Holberg:

3]. *Analog Design for CMOS VLSI Systems* by Franco Maloberti

4]. *Introduction to CMOS OP-AMPs and Comparators* by Roubik Gregorian:

5]. Start with a classic analog IC design textbook such as:

6]. Design of Analog CMOS Integrated Circuits by Behzad Raz